Predicting Daily S&P 500 Movements using Support Vector Machines and Forward Volatility

Abstract

Modelling and predicting the movements of financial instruments has always been a widely researched area in finance. This study focuses on using a branch of machine learning (Support Vector Machines) to forecast the daily price movements of the S&P 500 using a forward volatility feature, namely the VIX index. The predictions of the SVM are backtested across 27 years of data in order to quantify the effects that historical market scenarios have on the SVM accuracy. Various optimisation techniques are discussed and compared on the basis of speed and accuracy in order to determine the most effective optimisation methods. Once optimised, the effect of the VIX feature on predictive performance is analysed. Previous studies state that the inclusion of the VIX feature improves the accuracy of the SVM in bear markets when volatility is high. This research finds agreement with this statement in some instances but, in general, the inclusion of the VIX feature tends to cause the SVM to underperform when compared to an SVM without the VIX feature. The disagreement with other research is due to a number of shortcomings in previous studies. The subsequent analysis is of important when applying SVMs to financial time series in a.

Keywords

Support Vector Machines; Equity Index; Forecast; Volatility

1. Introduction

The efficient market hypothesis supposes that the current price of an index reflects all present information and, as such, must follow a Markovian process (Fama 1970). That is, any future evolution of the index price is dependant solely on its current state and past events have no influence on future index evolution. However, studies have shown that a varying degree of autocorrelation can exist in index price evolution, particularly in high frequency trading. This autocorrelation reduces with the frequency of trading but is still persistent in daily time intervals (Cont 2001). Using machine learning techniques, this research exploits autocorrelation present in the S&P 500 index to predict daily directional changes in its evolution i.e. to predict whether an upward or downward movement occurs. This study also incorporates historical values of the VIX index into the machine-learning algorithm to account for the effect of the prevailing market volatility on the S&P 500 evolution. As the VIX index is the market's expectation of future volatility, and volatility is one of the main drivers of autocorrelation (Bollerslev 1986), we test its ability to improve the accuracy of the machine learning algorithm.

The ability to extract profit from the financial markets is an obvious target for machine-learning but, by definition, capital markets follow a time-series of data and exploiting financial data requires a deft touch in terms of selecting the appropriate independent data points. A further challenge is the computational time required to extract decisions. This research examines the application of SVMs on daily S&P500 data. The incorporation of forward looking VIX data is a appealing addition and we examine the efficacy of its inclusion as an explanatory variable. We also look at computational times and put forward a new parameter search algorithm. This could have application as traders, particularly hedge funds, deploy SVMs from a daily to intra-day basis.

Applying machine learning techniques to financial markets has increased in popularity over the years due to the increase in computational power available. Machine learning techniques have been used in a variety of financial applications such as bankruptcy prediction (Min and Lee (2005); Shin *et al.* (2005)), credit card fraud detection (Bhattacharyya *et al.* 2011) and sentiment analysis (Smailović *et al.* (2013); Schumaker *et al.* (2012)), to name but a few. Support Vector Machines (SVMs) are a branch of machine learning which have also been applied to the financial markets. They are typically used for classification and regression purposes, and are particularly useful for analysing variables that have a binary outcome – such as the daily direction of the S&P 500, which is either 'up' or 'down'. The increased popularity of computer based trading has meant that a wide variety of SVM models have been developed to predict the future levels of financial instruments. This research uses a straight forward SVM methodology combined a forward-looking volatility-based feature[1] (the VIX index) to predict the future direction of daily S&P 500 movements. This methodology is applied across a 27-year time span using a rolling testing window, with and without the VIX feature, so the VIX feature's effect on SVM accuracy can be quantified. Algorithmic trading strategies, whose

[1] 'Feature' is the term used in machine learning to describe an individual measurable property or characteristic of the variable being observed.

buy/sell signals are generated by the SVM predictions, are also backtested and benchmarked against a simple buy and hold strategy to demonstrate the merits of the SVM algorithm.

SVMs are a branch of machine learning algorithm that are widely used in pattern recognition software and have been used to predict the level of financial instruments with varying degrees of success. Vapnik et al. (1997) first developed the support vector method to identify patterns in signals and to predict where the signal would move next. This process has since been applied to the financial markets, where instead of predicting signal movements, the price levels of financial instruments have been predicted. Various academic papers have illustrated the success in using SVMs for predicting index price movements. Most notably, Huang et al. (2005) show a 73% success rate in predicting weekly movements of the NIKKEI 225 index. Similar results are also obtained by Kara et al. (2011), where a 71.5% success rate is achieved in predicting daily ISE 100 index (Istanbul Stock Exchange) movements. These success rates are at the extreme end of the scale but serve to illustrate just how effective SVMs are under certain market conditions - both of these studies use a financial time series with strong directional trends. In Kara et al. (2011), the daily ISE index levels grew around 2400% in the testing time frame showing a strong upward trend, and, in Huang et al. (2005), the weekly NIKKEI 225 dropped around 80% over the testing time frame showing a strong downward trend. SVMs are suited to predicting movements in trending markets so these high accuracies come as no surprise. Unlike the two aforementioned studies, this research demonstrates the effect of different market trends (or lack of market trends in some cases) by using a 27-year time frame with a rolling testing window. The current literature on SVMs shows an average accuracy of 55%-60% is to be expected. It is worth noting however, that this average comes from research that incorporates a wide variety of financial instruments, prevailing market conditions and different SVM methodologies - see Van Gestel *et al.* (2001); Kim (2003); Choudhry and Garg (2008).

This research takes a novel approach in using a forward-looking volatility-based feature, CBOE's VIX index, to improve the accuracy of the SVM's predictions by accounting for the changing volatility environment. Volatility clustering is an empirical stylised fact observed in equity markets (Mandelbrot 1967), whereby volatility spikes tend to occur in clusters, and this clustering can also increase the level of autocorrelation observed in index movements (Bollerslev 1986). Since SVMs use index-dependant features such as lagged returns to train the SVM, and volatility and index levels are intrinsically linked, this auto-correlation can affect the accuracy of SVMs during periods of volatility clustering. There are reasons to suggest that volatility clustering could be both good and bad for the prediction accuracy of the SVM. For example, if index movements are more autocorrelated during periods of volatility clustering, and SVMs can recognize patterns caused by this autocorrelation, then future index movements could be more predictable to a well-trained SVM. However, increased volatility is, by its nature, an unpredictable influence so this may counteract any patterns recognised by the SVM.

Along with the VIX index, this research uses a variety of common technical indicators such as RSI, MACD and lagged returns as input features – these are all explained in the *Data* section. These features are all dependant on the historical closing price and serve to capture index movement trends such as momentum (i.e. rate of change of index level) and

convergence/divergence from the moving average. They are used with and without the VIX feature to demonstrate the effect of the VIX feature on prediction accuracy. A similar study by Rosillo *et al.* (2014) finds that SVMs using a VIX feature to predict weekly changes in the S&P 500 perform better than those that do not use a VIX feature, particularly in bear markets when volatility is high. However, some shortcomings in their methodology are noted. This study applies a similar methodology except on a daily interval across multiple market scenarios in a 27-year period. Also, to account for the perceived shortcomings in previous research, this study delves deeper into the SVM parameter optimisation process. This is a vital part of the model development as the performance of the SVM depends largely on the SVM input parameters. Once optimised, the SVM trading strategies are backtested across a range of historical market scenarios. This is particularly relevant in today's financial markets, where there is an ever increasing move towards algorithmic trading strategies. These strategies have to be capable of functioning across a variety of market scenarios and the VIX feature should help the SVM adapt to these different scenarios. In general, the algorithmic strategies developed over the course of this study show some promising results, but there is some disagreement with current literature on the effectiveness of the VIX feature.

Some similar studies analysing the effect of volatility on SVMs have been undertaken in the past. Yang et al. (2002) utilises a hybrid SVM to predict the movements of the Hang Sang index during periods of higher volatility. However, this method dynamically alters the SVM formula in order to adapt to the new volatility regime. This research differs from the aforementioned study by using a simpler form of SVM and including the VIX feature to adapt to different volatility regimes. Some dynamic adaptation of the input parameters is used in this research, however it is done so in a relatively simpler manner. This not only cuts down on computational time but is much more parsimonious than more complex hybrid models. A VIX feature has been integrated into the SVM algorithm in one previous study. Rosillo *et al.* (2014) use a VIX feature to adapt to different volatility regimes when predicting weekly changes in the S&P 500. They state that the prediction accuracy of SVMs improves when the VIX feature is used, particularly in bear markets with high volatility. It should be noted however, that Rosillo *et al.* (2014) use constant input parameters when using SVMs with different features. As is demonstrated later in the *Analysis* section, this is not appropriate as SVMs with different features require different input parameters to run optimally. So there are some reservations about the veracity of their results. That said, this research aims to emulate the results of Rosillo *et al.* (2014) except on a daily interval with a much larger testing window. The larger window is particularly important as the effect of different market scenarios on SVM accuracy can be better quantified, rather than just analysing favourable trending markets. The perceived shortcomings in the research by Rosillo *et al.* (2014) is also corrected by optimising the SVMs independently for each new feature space.

This rest of this paper is organised as follows; the next section, methodology, details some background theory on SVMs and how they are optimised is also provided, along with a description of the test cases used in this study. The next data section provides detail on the sources of all data, and how the data is pre-processed before being input into the SVM. An explanation of the various features used in training the SVM is also provided. After this, a

detailed discussion on the optimisation techniques, testing and performance of the SVM is presented. Concluding remarks are given in the final section.

2. Methodology

2.1 Optimising SVM Parameters

The optimum choice of parameters, C and γ, is vital to the predictive performance of the SVM. There is no agreed solution to finding the correct combination of these parameters so time-consuming optimisation techniques must be used. Each dataset has its own unique optimum parameter combination, which means the SVM must be continually updated with the arrival of new inputs. This a reasonably well studied area and so researchers have developed multiple methods to expedite the process of finding the optimum parameters. In total, 6 automated methods are tested in this study. This comprises of 2 optimisation algorithms each used in combination with 3 automated search methods. The two optimisation algorithms are: Sequential Minimal Optimisation (SMO) (Platt 1998) and Iterative Single Data Algorithm (ISDA) (Kecman et al. 2005) - more detail on these algorithms can be found in the referenced papers. The 3 search methods are: 'Grid search', which essentially creates a grid of C and γ values and runs the SVM with every combination in the grid. 'Random search' is essentially the same as 'Grid search' except the combination of parameters is chosen at random, rather than in sequential order, so it can potentially converge on an optimum solution faster. 'Bayesian Optimisation' uses a Bayesian strategy to estimate parameter combinations given a randomised initial prior probability and random initial combination of C and γ. The prior is updated continually as sampling progresses in order to converge on an optimal solution. All of these methods find the optimal solution by minimising the 'loss function' of the SVM. This function is used widely in machine learning and essentially describes the magnitude of the classification mistakes that the SVM made. This means the optimisation methods aim to produce an SVM that has the lowest potential to make large mistakes. A comparison of each search method and optimisation routine is presented in the Calculation section to demonstrate which is the most effective in terms of computational time and accuracy.

One problem that exists with the automated optimisation methods is that they can be computationally intensive, which is a major drawback for the large test cases in this study. With this in mind, for this article we devise a method to optimise the SVM in a faster time but with some trade off in out of sample accuracy – we term it the Proximity Grid Search Algorithm. This new method is essentially the same as the grid search technique described before except the optimal solution is found by maximising the in-sample accuracy. Firstly, a course grid search is carried out on the parameter space to find an area that performs well for the first testing window. This is done by varying the values of C and γ across the range $\{2^{-10}, 2^{-9} \ldots 2^{9}, 2^{10}\}$ (as recommended in Hsu *et al.* (2008)) and locating an area on the grid which yielded the highest in-sample accuracy. A more localised search can be carried out around this portion of the grid to improve the accuracy further if desired. This is the starting parameter combination for the proximity grid search algorithm. As the testing window rolls forward by a relatively small time frame in relation to the overall window size, a lot of the data overlaps and so the optimum parameter combination will not change by a significant amount. Using the

previous window's optimum parameter combination as the starting point, a small grid search is carried out on the grid region by discretising the area around the optimum parameters into a 5x5 grid. This gives 25 combinations of parameters that may (or may not) be better than the previously obtained values – see Figure 1 for illustration. Once an optimum parameter combination for this window is found, the SVM is run with these parameters, the testing window is then rolled forward, and the process repeated. The optimum parameters do not deviate much due to the overlap of data, which allows the optimisation process to track the new data in the rolling window, rather than starting from anew every time. It is worth noting that if equal maximum in-sample accuracies are obtained for multiple grid points, the inner grid and centre point are given preference as the new optimum point. This is to prevent the grid moving too quickly across the parameter space, which will destabilise the solution. Due to the smaller size of the grid in the proximity grid search algorithm, it is much faster than the automated optimisation routines but has some trade off in out of sample accuracy.

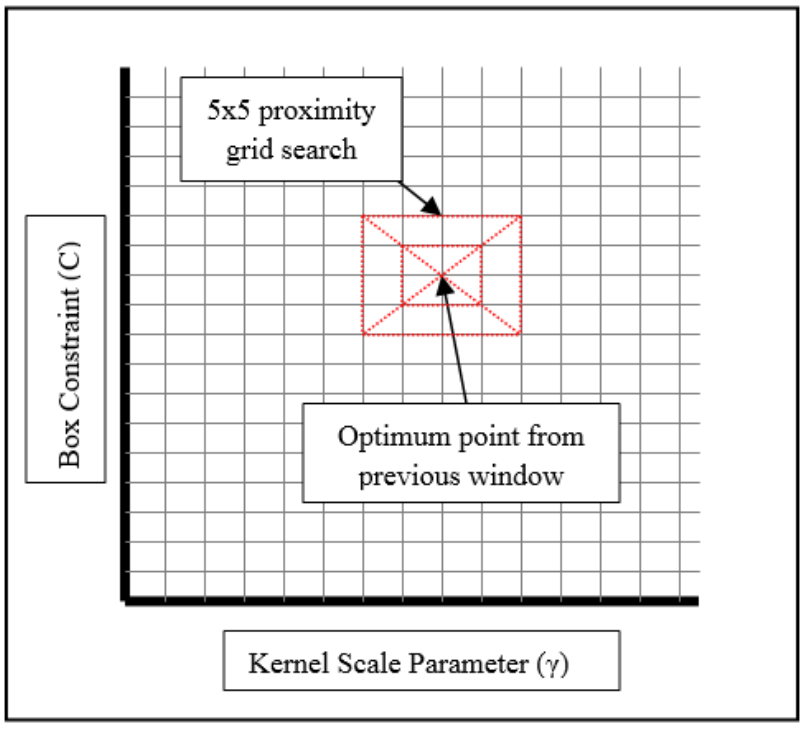


Figure 1: Localised grid search of the SVM parameter space

2.2: Test Cases

In some of our test cases, isolated historical periods such as the Dotcom Crash of the early 2000's are used to demonstrate the performance of the SVM during particular market scenarios. All of these scenarios are run with and without the VIX feature to demonstrate how the inclusion of the VIX feature affects the SVM performance. However, for the majority of the analysis, the SVM is run across a 27-year timeframe, from 13/02/1990 to 17/02/2017, using a rolling testing window. The aim of this is to demonstrate the effect of prevailing market conditions on prediction accuracy. The size of the rolling sampling window is also varied to quantify the effect of window size on predictive performance. A schematic of how the rolling window operates is given in Figure 2. Here it can be seen that the testing window simply rolls

forward in steps of *n*, which is the size of the out of sample window size. It shows that the testing window rolls forward, in increments equal to the out of sample window size, until the out the sample set reaches the end of the data.

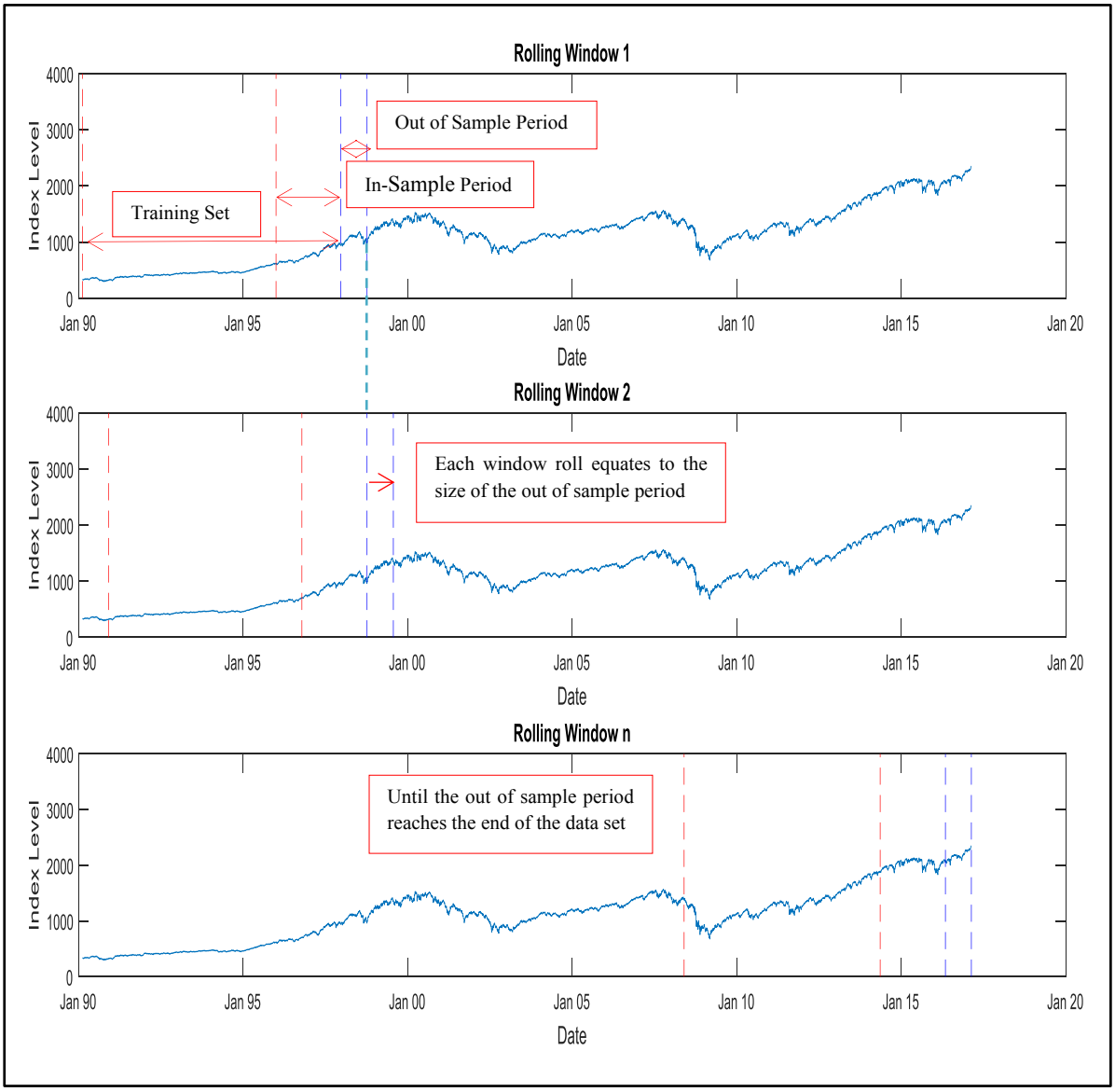


Figure 2: How the rolling testing window operates

In each testing window, the SVM is trained using the 'training set' and the in-sample accuracy is obtained by validating the SVM against the last year of the training set. This is what is known as 'holdout validation' and is much simpler than the more common approach of using cross validation in SVMs. Cross validation is generally not appropriate for use with financial time series as this type of validation does not take account of the time-dependant nature of a financial time series. The in-sample accuracy measures how well trained the SVM is and gives an indication of its future predictive performance. The trained SVM is then applied to the next *n*-days preceding the training set to measure the out of sample accuracy. The out of sample size is n=30 days in all test cases to coincide with CBOE's VIX index methodology - the VIX index is based on the forward price of S&P 500 options 30-days ahead.

2.3 Performance Measurement

The main metric used to compare the various SVM configurations and testing periods is simply a percentage score of number of correct predictions over the total number of observations. This is done for the out of sample and in-sample testing windows. In some cases, the time taken to run the SVM model is discussed. Graphical representations of how the accuracies (in-sample and out of sample) evolve with time are also presented. This is to judge the effect historical market scenarios have on the SVM accuracy. To conclude, the P&Ls of backtested SVM trading strategies are analysed so that the benefits of the VIX feature can be quantified.

3. Data and Calculation

3.1: Data Collection and Pre-Processing

The daily closing price for the S&P 500 and daily closing level of the VIX index for the dates 13/02/1990 to 17/02/2017 were downloaded from Bloomberg. The closing price of the S&P 500 is used to calculate all the technical indicators and returns data that are needed to train the SVM. A flag for an upward or downward movement is also placed into the data set to classify whether the index level moved up or down from day to day. This is the response variable that the SVM predicts. As per Rosillo *et al.* (2014), the SVM features are lagged at 1-3 days and placed into the data set. The data is normalised to ensures that larger variables do not overwhelm the system.

3.2: SVM Features

Despite some scepticism in academic finance, technical analysis (Park and Irwin 2007) is widely used today by fund managers. Menkhoff (2010) surveyed 692 fund managers across 5 countries (US, Switzerland, Germany, Italy, and Thailand) and found that 87% of fund managers use technical analysis to some extent. Interestingly, the fund managers surveyed believe that one of the major benefits of technical analysis is that it allowed them to gauge the current psychology of the market. Given the majority of market participants act in a heterogeneous manner to market trends, technical analysis allows them to see how the market is reacting to different technical indicators. From this they can gauge the 'mood' of the market and adjust their positions accordingly. Although not based on any quantitative theory, the fact that many market participants use these indicators to generate buy and sell signals means there is information that can be gleaned from them. We submit that SVMs can then spot trends in how the market behaves given past values of these technical indicators and then predict the future direction of the S&P 500 based on this trend.

Kim (2003) shows a wide array of features that can be used to train the SVM, however, our research uses 4 features that cover a range of technical traits. This include lagged returns (LRs), Moving Average Convergence Divergence (MACD), Relative Strength Index (RSI) and VIX. The MACD and RSI are chosen as they are commonly used indicators by technical traders. LRs are included as they account for the autocorrelation in the S&P 500 time series. The VIX index is used to account for the volatility of the market. Lags of 1-3 days are used as per Rosillo *et al.* (2014).

Relative Strength Index (RSI) was formulated by Wilder Jr. (1978), and is a technical indicator that is often referred to as a 'momentum oscillator'. This means it is represents the tendency of an index to have large day to day movements. High RSI suggests that the current trend is likely to have large upward jumps. Low RSI purportedly signal when large downward jumps are likely to occur. The momentum of the price changes is calculated by the ratio of higher closes to lower closes over a certain time frame. This research uses a 14-day time frame as it is often used as the default time period by technical traders. It is calculated by equations 2 & 3:

$$RSI = 100 - \frac{100}{1+RS} \qquad (2)$$

Where:

$$RS = \frac{SMMA_{U,n}}{SMMA_{D,n}} \qquad (3)$$

The relative strength factor (RS) is measured by the ratio of the exponentially smoothed moving average (SMMA) of upwards and downwards price changes over the n-day period. RSI has been used as a feature in SVMs by other researchers in the past – see: Kim (2003); Huang and Tsai (2009); Rosillo *et al.* (2014).

Moving Average Convergence Divergence (MACD) is technical indicator used by traders to measure the direction and duration of an index movement trend. It is created by getting the difference between the exponential moving average (EMA) of the index price over a short period (12-day) and a longer period (26-day). The shorter period EMA is more sensitive to the current market conditions than the longer period EMA so the changing values of the MACD series can signify the beginning (or end) of a market trend is about to occur. This is particularly useful as downturns are often coupled with increased volatility and can help the SVM improve predictions in such unpredictable scenarios. The MACD has been successfully used as a feature in previous studies such as Cao and Tay (2001b); Kumar and Thenmozhi (2006); Kara *et al.* (2011).

The VIX index is an implied volatility index. It measures the market's expectation of 30-day volatility implicit in the prices of near-term S&P 500 options (CBOE 2015). The VIX index is calculated by obtaining the 30-day standard deviation of the forward price of a strip of options. The forward price of this strip is calculated at each minute of the 30-day period for increased accuracy. The strip of options is made up of OTM calls and puts to account for the skew that is present in the S&P 500 implied volatility. Volatility is caused by the activity of market participants, then this index describes the expected future volatility of the S&P 500. In other words, the expected future behaviour of market participants is inherent in the VIX level. There are some characteristic traits of the VIX index which make it particularly suitable for use with pattern recognition algorithms. Typically, the VIX index has an asymmetric inverse relationship with the S&P 500 in that it is much more sensitive to downward movements of the S&P 500 than it is to upward movements. Another property of the VIX index is that it follows an Ornstein-Uhlenbeck process, which means it tends to revert to long run mean over time.

3.2: Calculation

We first deploy the SVM in a bear market with high volatility, with and without the VIX feature to replicate previous research by Rosillo *et al.* (2014). We then present a comparison of our proximity grid search algorithm (PGSA) against other automated optimisation routines. Finally, we examine the performance of the SVM with and without the VIX feature using the rolling testing window.

3.2.1: Use of VIX Feature in a Bear Market

Rosillo *et al.* (2014) state that the inclusion of VIX in the SVM feature space improves the accuracy of the SVM when predicting weekly S&P 500 direction changes, particularly in bear markets when volatility is high. To prove this concept on a daily time frame, the SVM is applied to a historical period where there is a clear bear market and high volatility. A historical period where this type of market prevailed would be the Dotcom Crash, which lasted approximately from March 2000 until Oct 2002. The timeframe in question is shown below in Figure 3, where a clear bear trend is observed in the S&P 500 and spikes of high volatility observed in the VIX index.

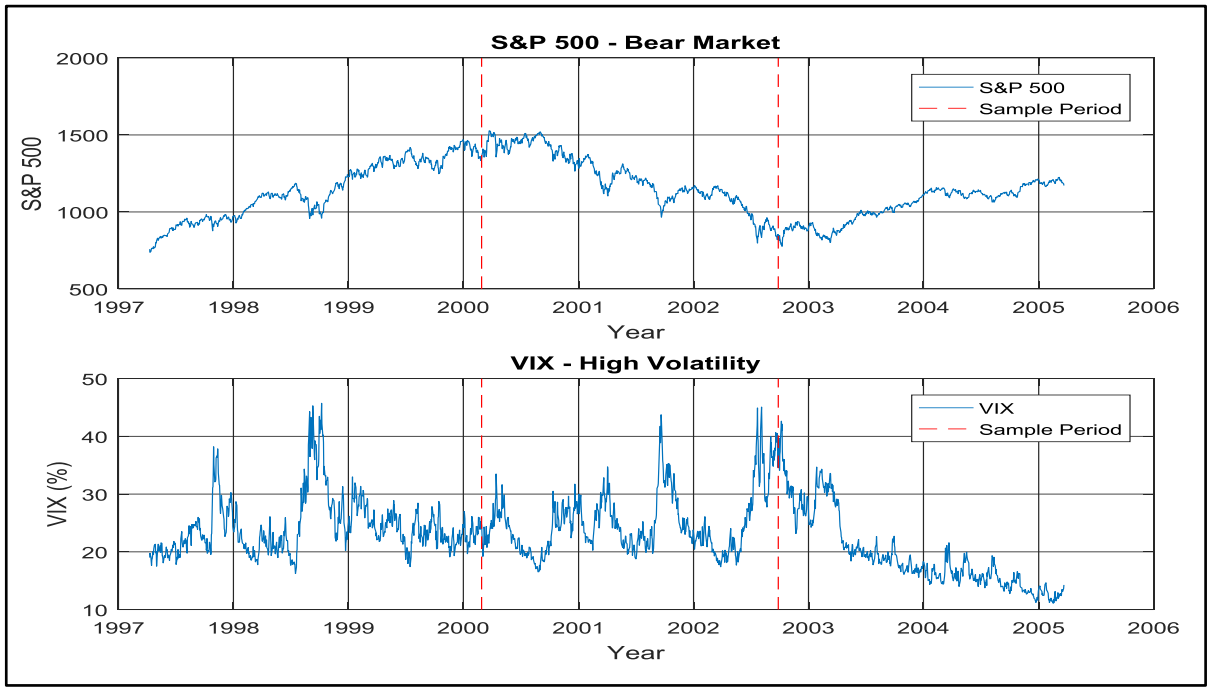


Figure 3: Historical levels of the S&P 500 and VIX for the Dotcom Crash

This time frame is analysed with and without the VIX feature. The out of sample period is the last 30 days of the data set, and the in-sample accuracy is validated against the last 80 days of the training set. The results are presented in Table 1: Automated optimisation results with and without VIX feature. Must mention note on computation time. The search methods are constrained in the bounds of C = [20, 90] and γ = [0.15, 2] to prevent over/underfitting. See Cao & Tay (2003) for a more detailed discussion.

	SMO - Gridsearch					ISDA - Gridsearch				
	In Sample Accuracy (%)	Out of Sample Accuracy (%)	C	Gamma	Time Taken (sec)	In Sample Accuracy (%)	Out of Sample Accuracy (%)	C	Gamma	Time Taken (sec)
With VIX	58.33	40	39	1.5	92	58.33	50	54.5	0.2	346
Without VIX	56.94	53.33	20	1.125	120	56.94	53.33	20	1.125	388

	SMO - Random					ISDA - Random				
	In Sample Accuracy (%)	Out of Sample Accuracy (%)	C	Gamma	Time Taken (sec)	In Sample Accuracy (%)	Out of Sample Accuracy (%)	C	Gamma	Time Taken (sec)
With VIX	58.33	43.33	36.54	0.151	27	58.33	43.33	36.54	0.151	104
Without VIX	56.94	53.33	34.7	1.125	26	56.94	53.33	34.7	1.125	127

	SMO - Bayesian					ISDA - Bayesian				
	In Sample Accuracy (%)	Out of Sample Accuracy (%)	C	Gamma	Time Taken (sec)	In Sample Accuracy (%)	Out of Sample Accuracy (%)	C	Gamma	Time Taken (sec)
With VIX	58.33	43.33	89.87	0.15	77	58.33	40	89.2	0.156	178
Without VIX	56.94	53.33	33.42	1.091	71	56.94	56.66	20.4	1.997	172

Table 1: Automated optimisation results with and without VIX feature. Must mention note on computation time

In table 2, the inclusion of the VIX feature increases the in-sample accuracy achieved by the SVM for all optimisations. Conversely however, the out of sample accuracy is lower for all SVMs with the VIX feature – this is a direct contrast to what would have been expected given the previous research by Rosillo *et al.* (2014). To understand this phenomena, we examined the accuracy of the SVM with and without the VIX and for both in-sample and out-of-sample. We found including the VIX tends to reduce the overall Cost/γ combinations to a specific answer. In other words, when the VIX is introduced, overfitting is promoted. This could be due to some idiosyncrasy of this particular sample period. So to account for any potential idiosyncrasies, the SVMs is applied to a rolling testing window in Section 3.3 where all stages of the Dotcom crash are incorporated into the out of sample testing.

3.2.2: Computing Optimum Parameters

The results in Table 1 show that all optimisation methods converged on the same in-sample accuracies for SVMs with or without the VIX feature. This is due to the fact that the algorithms do not seek to maximise prediction accuracy but rather minimise the loss function of the SVM. Due the variance associated with the optimum values for C and γ, automated search routines are computationally intensive so we developed the proximity grid search algorithm (PGSA) was developed to expedite the process. We present the results in Table 2 below where we compare both in-sample and out of sample results for varying window sizes. The SMO–Bayesian SVM finishes with a higher out of sample accuracy than the PGSA SVM by 1.18%. Clearly maximising the in-sample accuracy does not necessarily result in better predicative performance in the out of sample testing. However, that the proximity grid search algorithm runs much quicker than the SMO – Bayesian method, this is not an insignificant factor, particularly for intra-day traders.

	2 year rolling window					
	Proximity Grid Search 5x5			SMO - Bayesian		
	Average In Sample Accuracy (%)	Average Out of Sample Accuracy (%)	Time Taken (minutes)	Average In Sample Accuracy (%)	Average Out of Sample Accuracy (%)	Time Taken (minutes)
With VIX	54.51	52.2	15	53.32	53.38	299
Wiithout VIX	53.22	51.59	33	53.32	53.38	283

	5 year rolling window					
	Proximity Grid Search 5x5			SMO - Bayesian		
	Average In Sample Accuracy (%)	Average Out of Sample Accuracy (%)	Time Taken (minutes)	Average In Sample Accuracy (%)	Average Out of Sample Accuracy (%)	Time Taken (minutes)
With VIX	54	51.76	68	52.37	52.36	322
Wiithout VIX	54.26	51.78	95	52.3	52.14	299

	8 year rolling window					
	Proximity Grid Search 5x5			SMO - Bayesian		
	Average In Sample Accuracy (%)	Average Out of Sample Accuracy (%)	Time Taken (minutes)	Average In Sample Accuracy (%)	Average Out of Sample Accuracy (%)	Time Taken (minutes)
With VIX	53.15	53	251	52.21	53.33	386
Wiithout VIX	52.58	52.81	226	52.59	53.1	410

	10 year rolling window					
	Proximity Grid Search 5x5			SMO - Bayesian		
	Average In Sample Accuracy (%)	Average Out of Sample Accuracy (%)	Time Taken (minutes)	Average In Sample Accuracy (%)	Average Out of Sample Accuracy (%)	Time Taken (minutes)
With VIX	54.14	51.37	307	52.73	52.96	490
Wiithout VIX	54.72	52.33	237	52.93	52.94	464

Table 2: Results with & without VIX, compared by window size and search method

Larger sample periods can provide the SVM with more information about how the S&P 500 reacts to different market scenarios, and so should improve overall accuracy when the index level is more erratic. Conversely, SVMs with smaller sample periods have the benefit of a certain 'naivety' when the market is in a clear trend. They only have information on the current market trend and so will predict the continuation of that trend. In such scenarios, this results in a higher accuracy for smaller sample periods – that is until the market trend ends. The most accurate and stable solutions come from the 2 and 8 year windows but it should be noted that the final out of sample accuracy average across all 4 window sizes only differs by 1.02% between the best and worst result.

3.3: Analysis of VIX feature

Using the SMO-Bayesian method of optimisation and a 2-year sample period, the use of the VIX feature is analysed across the 27-year period. The predictions from both SVM models (one with the VIX feature, one without the VIX feature) are used to generate daily buy/sell signals for an algorithmic trading strategy. The performance of the SVM strategies is benchmarked against a buy and hold strategy, and the results are shown below in Figure 4.

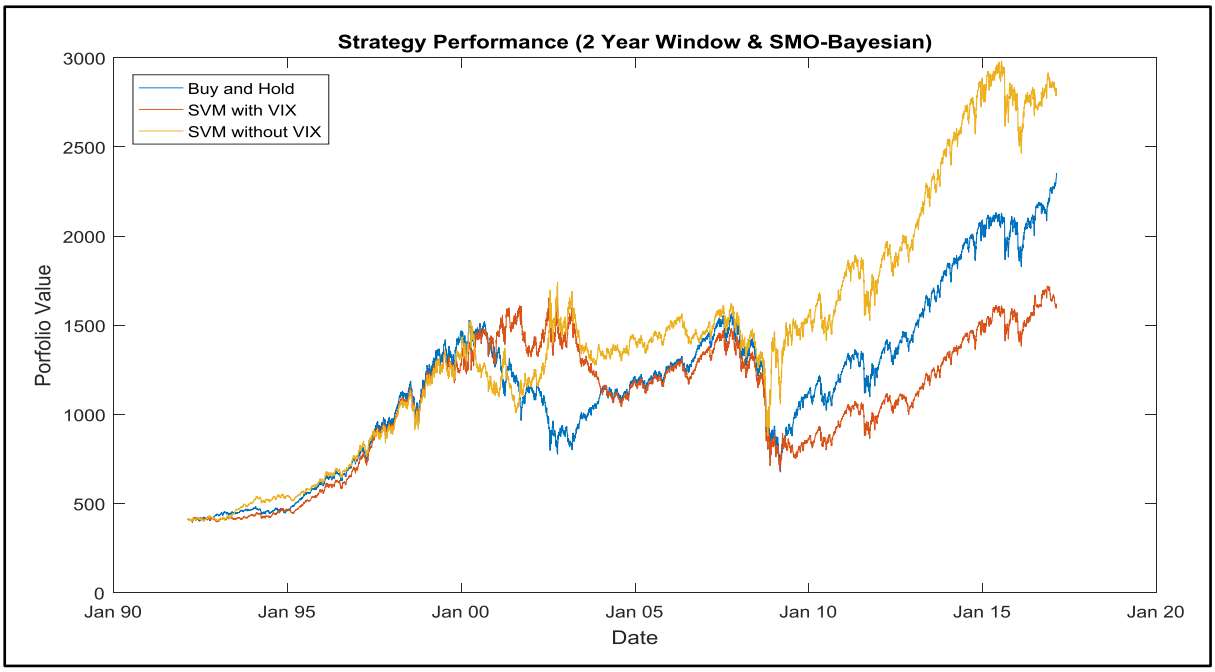


Figure 4: Strategy performance using 2-year window and SMO-Bayesian method

It can be seen that the SVM with the VIX feature outperforms the SVM without the VIX feature for the majority of the Dotcom crash (2000-2002) as observed by Rosillo *et al.* (2014) on a weekly time interval. Figure 4 also agrees with previous observations where the SVM without the VIX feature outperformed the SVM with the VIX feature in the final period of the Dotcom crash. However, the period during the great recession does not show a similar trend to the Dotcom crash, which is unexpected given both these periods are bear markets with high volatility. One reason for this is market timing, the SVM with VIX correctly predicted some of the largest downward moves towards the end of the Financial Crisis in 2009 and made a profit from this movement when the other 2 strategies did not. It is clear that after the financial crisis all 3 strategies follow a similar.

That said, a notable trend in Figure 4 is that the SVM without the VIX feature outperforms the SVM with the VIX feature when the market transitions from a bear market to a bull market (around Q4 2002 and Q2 2009) and maintains a better performance during the ensuing bull markets. In fact, in the initial bull market from 1992 – 2000, the SVM without the VIX feature outperforms the SVM with the VIX feature as well. Indeed, in all 3 major bull markets in the sample data, the SVM without the VIX feature performs better than the SVM with the VIX feature. The 2 bear markets present in the sample data show conflicting results. While there is some agreement with Rosillo *et al.* (2014) during the Dotcom crash in that the SVM with the VIX feature performs better, overall the VIX appears to confuse the SVM somewhat and lead to worse results in general. This is more than likely due to the fact that Rosillo *et al.* (2014) used constant parameters for the SVMs with and without the VIX feature. We feel that the two models should be optimised independently. In general, there is no significant benefit from including the VIX feature in the SVM feature space although, it did show some promise during some stages of the Dotcom Crash. With this in mind, the VIX feature could possibly be

implemented into the SVM via a dynamic feature selection algorithm when certain market scenarios arise (e.g. Lee (2009); Huang and Tsai (2009)).

4. Conclusion

This research proves that some of the past studies on the inclusion of a VIX feature in the SVM algorithm make assumptions which may have led to inaccurate conclusions. A more correct approach is used in this research whereby the SVMs are independently optimised at each sample period to find the optimum parameter combination. Using this method, it can be seen that, in general, no significant benefit is gained from including the VIX feature. That said, some benefit is noted during the Dotcom Crash, which means the VIX feature could be incorporated to the SVM algorithm on a dynamic basis. However, more research would need to be carried out on this. In general, the VIX feature appears to confuse the SVM around periods of market transition from a bear market to a bull market. It also causes the SVM to underperform during bull markets. The model with fewer features (SVM without VIX) performed consistently better than the SVM with the VIX feature. This is possibly due to the fact that financial time series can be quite noisy and this noisiness makes it difficult for SVMs to find patterns in the data. The VIX index itself is also a relatively noisy time series given its large trading volume. The inclusion of another noisy time series at 3 lags appears to make the SVM underperform in most cases. There is definitely merit in using SVMs to predict the future movements of financial instruments, but the use of the VIX feature to enhance the performance of the SVM does not appear to consistently work. Its noisy nature further compounds the difficulties in spotting trends in an already noisy time series. Further research should be carried out on possibly including different (and less noisy) features in the SVM algorithm. Only 4 features at lags of 1-3 days (12 in total) are used to achieve an average accuracy of around 53%. However, other studies such as Kim (2003) used up to 12 features at different lags. This is one of the earlier studies on SVMs in financial forecasting so there is some degree of overkill on the amount of features. Nevertheless, a more detailed study on the use of various macroeconomic and technical indicators should be carried out to see which features yield the best results.

Another noteworthy point is how important the optimisation process is to the SVM performance. Multiple optimisation methods have been tested and the SMO algorithm with a Bayesian search method was found to yield the best out of sample accuracy. However, the long runtimes of this method make it difficult to implement into shorter frequency trading strategies. That said, attempts to expedite the parameter selection process result in worse out of sample accuracies. The proximity grid search algorithm produces significantly reduced runtimes but lower average out of sample accuracy (on average 0.833% lower across all test cases). This may seem like a small difference, but the fact that the average out of sample accuracy for the Bayesian-SMO method is only at 52.95% means that it is only 2.95% away from a 50/50 guess. The automated methods optimise the parameters by minimising the loss function of the SVM. This produces a better out of sample accuracy when compared to the proximity grid search algorithm, which optimises the parameters by maximising the in-sample accuracy. Further research on parameter selection could also be done by hybridising the models in this study to include research by Kazem *et al.* (2013) - who implement a 'chaotic firefly' algorithm to choose

the SVM parameters. This is much more efficient method than the SMO-Bayesian method used in this research but is relatively more complex to implement. Finally, a more detailed study on the effect of rolling window size could be carried out to see which yield the best results – although this research suggests smaller sample periods result in a more stable and accurate solution.

In summary, this research demonstrates the power of machine learning in predicting the movements of financial instruments. SVMs are just a single branch of machine learning algorithm and, as discussed in the *Literature Review*, there are numerous other successful uses of machine learning in financial forecasting. While the inclusion of a forward-looking volatility based indicator (the VIX index) does not generally improve the predictive performance of the SVM, this research still demonstrates how a relatively simple SVM model can be used to predict the movements of the S&P 500 on a daily time interval. In fact, the model with the least amount of features performed best in backtesting – showing how effective a simple SVM can be when compared to a buy and hold strategy. With the increase popularity of machine learning and SVMs, more software tools and packages are being developed to bring machine learning to a wider audience. This is bound to have a major impact on financial markets and studies already estimate that over 62% of hedge funds invest in machine learning to enhance their trading activities (Butcher 2017) - and this number is set to grow as the power of computers and the knowledge of machine learning increases in the coming years.

5. References

Ballings, M., Van Den Poel, D., Hespeels, N., Gryp, R. (2015) 'Evaluating multiple classifiers for stock price direction prediction', *Expert Systems with Applications*, 42(20), 7046–7056.

Bhattacharyya, S., Jha, S., Tharakunnel, K., Westland, J.C. (2011) 'Data mining for credit card fraud: A comparative study', *Decision Support Systems*, 50(3), 602–613.

Black, F., Scholes, M. (1973) 'The Pricing of Options and Corporate Liabilities', *The Journal of Political Economy*, 81(3), 637–654.

Bollerslev, T. (1986) 'Generalized autoregressive conditional heteroskedasticity', *Journal of Econometrics*, 31(3), 307–327.

Burges, C.J.C. (1998) 'A tutorial on support vector machines for pattern recognition', *Data Mining and Knowledge Discovery*, 2(2), 121–167.

Butcher, D. (2017) Barclays' Guide to AI and Big Data Jobs in Hedge Funds [online], *eFinanceCareers*, available: http://news.efinancialcareers.com/ie-en/287300/barclays-machine-learning-ai-and-big-data-jobs-in-hedge-funds/ [accessed 9 Aug 2017].

Cao, L., Tay, F.E.H. (2001a) 'Application of support vector machines in financial time series forecasting', *Omega*, 29(4), 309–317.

Cao, L., Tay, F.E.H. (2001b) 'Financial Forecasting Using Support Vector Machines', *Neurocomputing*, 1(2), 1–36.

Cao, L., Tay, F.E.H. (2003) 'Support vector machine with adaptive parameters in financial time series forecasting.', *IEEE Transaction on Neural Networks*, 14(6), 1506–18.

CBOE (2015) 'The CBOE Volatility Index - VIX', *CBOE White Paper*, 1–23.

Chang, C.-C., Lin, C.-J. (2013) 'LIBSVM : A Library for Support Vector Machines', *ACM Transactions on Intelligent Systems and Technology (TIST)*, 2, 1–39.

Chen, W.-H., Shih, J.-Y., Wu, S. (2006) 'Comparison of support-vector machines and back propagation neural networks in forecasting the six major Asian stock markets', *Int. J. Electronic Finance*, 1(1), 49–67.

Choudhry, R., Garg, K. (2008) 'A Hybrid Machine Learning System for Stock Market Forecasting', *World Academy of Science, Engineering and Technology*, 2(15), 315–318.

Cont, R. (2001) 'Empirical properties of asset returns: stylized facts and statistical issues', *Quantitative Finance*, 1(April), 223–236.

Fama, E.F. (1970) 'Efficient Capital Markets: A Review Of Theory And Empirical Work', *The Journal of Finance*, 25(2), 383–417.

Gavrishchaka, V. V., Banerjee, S. (2006) 'Support vector machine as an efficient framework for stock market volatility forecasting', *Computational Management Science*, 3(2), 147–160.

Van Gestel, T., Suykens, J.A.K., Baestaens, D.-E., Lambrechts, A., Lanckriet, G., Vandaele, B., De Moor, B., Vandewalle, J. (2001) 'Financial Time Series Prediction Using Least Squares Support Vector Machines Within the Evidence Framework', *IEEE Transaction on Neural Networks*, 12(4), 809–821.

Heston, S. (1993) 'A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options', *The Review of Financial Studies*, 6(2), 327–343.

Hsu, C.-W., Chang, C.-C., Lin, C.-J. (2008) 'A Practical Guide to Support Vector Classification', *BJU international*, 101(1), 1396–400.

Huang, C.-L., Tsai, C.-Y. (2009) 'A hybrid SOFM-SVR with a filter-based feature selection for stock market forecasting', *Expert Systems with Applications*, 36(2), 1529–1539.

Huang, W., Nakamoria, Y., Wang, S.-Y. (2005) 'Forecasting stock market movement direction with support vector machine', *Computers and Operations Research*, 32(10), 2513–2522.

Kacmajor, T. (2016) SVM Model Selection - How to Adjust All These Knobs Pt. 1 [online], *ProggBlogg*, available: http://tomaszkacmajor.pl/index.php/2016/04/24/svm-model-selection/ [accessed 7 Jun 2017].

Kara, Y., Acar Boyacioglu, M., Baykan, Ö.K. (2011) 'Predicting direction of stock price index movement using artificial neural networks and support vector machines: The sample of the Istanbul Stock Exchange', *Expert Systems with Applications*, 38(5), 5311–5319.

Kazem, A., Sharifi, E., Hussain, F.K., Saberi, M., Hussain, O.K. (2013) 'Support vector

regression with chaos-based firefly algorithm for stock market price forecasting', *Applied Soft Computing*, 13(2), 947–958.

Kecman, V., Huang, T., Vogt, M. (2005) 'Iterative Single Data Algorithm for Training Kernel Machines from Huge Data Sets: Theory and Performance', *Support Vector Machines: Theory and Applications*, 605–605.

Kim, K. (2003) 'Financial time series forecasting using support vector machines', *Neurocomputing*, 55(1–2), 307–319.

Kumar, M., Thenmozhi, M. (2006) 'Forecasting Stock Index Movement : a Comparision of Support Vector Machines and Random Forest', *Indian Institute of Capital Markets 9th Capital Markets Conference Paper.*, 1–16.

Lee, M.-C. (2009) 'Using support vector machine with a hybrid feature selection method to the stock trend prediction', *Expert Systems with Applications*, 36(8), 10896–10904.

Lu, C.-J., Lee, T.-S., Chiu, C.-C. (2009) 'Financial time series forecasting using independent component analysis and support vector regression', *Decision Support Systems*, 47(2), 115–125.

Mandelbrot, B. (1967) 'The Variation of Some Other Speculative Prices', *The Journal of Business*, 40(4), 393–413.

Menkhoff, L. (2010) 'The use of technical analysis by fund managers: International evidence.', *Journal of Banking and Finance*, 34(11), 2573–2586.

Min, J.H., Lee, Y.-C. (2005) 'Bankruptcy prediction using support vector machine with optimal choice of kernel function parameters', *Expert Systems with Applications*, 28(4), 603–614.

Ou, P., Wang, H. (2009) 'Prediction of Stock Market Index Movement by Ten Data Mining Techniques', *Modern Applied Science*, 3(12), 28–42.

Pai, P.-F., Lin, C.-S. (2005) 'A hybrid ARIMA and support vector machines model in stock price forecasting', *Omega*, 33, 497–505.

Park, C.-H., Irwin, S.H. (2007) 'What do we know about the profitability of technical analysis?', *Journal of Economic Surveys*, 21(4), 786–826.

Platt, J.C. (1998) 'Sequential Minimal Optimization: A Fast Algorithm for Training Support Vector Machines', *Advances in kernel methods*, 185–208.

Rosillo, R., Giner, J., de la Fuente, D. (2014) 'The effectiveness of the combined use of VIX and support vector machines on the prediction of S&P 500', *Neural Computing and Applications*, 25(2), 321–332.

Schumaker, R.P., Zhang, Y., Huang, C., Chen, H. (2012) 'Evaluating sentiment in fi nancial news articles', *Decision Support Systems*, 53(3), 458–464.

Shin, K.-S., Lee, T.S., Kim, H. (2005) 'An application of support vector machines in bankruptcy prediction model', *Expert Systems with Applications*, 28(1), 127–135.

Smailović, J., Grčar, M., Lavrač, N., Žnidaršič, M. (2013) 'Predictive sentiment analysis of

tweets: A stock market application', *Human-Computer Interaction and Knowledge Discovery in Complex, Unstructured, Big Data*, 1(1), 77–88.

Vapnik, V., Golowich, S.E., Smola, A. (1997) 'Support Vector Method for Function Approximation, Regression Estimation, and Signal Processing', *Advances in Neural Information Processing Systems 9*, 281–287.

Wilder Jr., J.W. (1978) *New Concepts in Technical Trading Systems*, Trend Research.

Yang, H., Chan, L., King, I. (2002) 'Support Vector Machine Regression for Volatile Stock Market Prediction', *Intelligent Data Engineering and Automated*, 391–396.